

UNIT – 2
ASSESSMENT – 3

PROBLEM

1. The A* Search Algorithm is an informed search algorithm used to find the shortest and most optimal path between a start node and a goal node in a weighted graph. It combines the actual path cost $g(n)$ and the heuristic estimate $h(n)$ to evaluate each node using $f(n) = g(n) + h(n)$. The objective is to determine the optimal path while minimizing the total travel cost.

2. The **Minimax Algorithm with Alpha-Beta Pruning** is an Artificial Intelligence search algorithm used in two-player, zero-sum games to determine the best possible move. The algorithm assumes that the **MAX** player tries to maximize the score, while the **MIN** player tries to minimize it. Since evaluating every possible game state is computationally expensive, **Alpha-Beta Pruning** eliminates branches that cannot influence the final decision, thereby improving search efficiency without affecting the optimal result.